



CLASSROOM CONTACT PROGRAMME

(Academic Session : 2021 - 2022)

JEE(Advanced)
FULL SYLLABUS
09-06-2022

JEE(Main + Advanced) : ENTHUSIAST COURSE : [SCORE-I (ALL STAR BATCH)]

PAPER-2

Time : 3 Hours

Maximum Marks : 198

READ THE INSTRUCTIONS CAREFULLY

GENERAL :

1. This sealed booklet is your Question Paper. Do not break the seal till you are told to do so.
2. Use the Optical Response sheet (ORS) provided separately for answering the questions.
3. Blank spaces are provided within this booklet for rough work.
4. Write your name, form number and sign in the space provided on the back cover of this booklet.
5. After breaking the seal of the booklet, verify that the booklet contains **24** pages and that all the **18** questions in each subject and along with the options are legible. If not, contact the invigilator for replacement of the booklet.
6. You are allowed to take away the Question Paper at the end of the examination.

OPTICAL RESPONSE SHEET :

7. The ORS will be collected by the invigilator at the end of the examination.
8. Do not tamper with or mutilate the ORS. **Do not use the ORS for rough work.**
9. Write your name, form number and sign with pen in the space provided for this purpose on the ORS. **Do not write any of these details anywhere else on the ORS.** Darken the appropriate bubble under each digit of your form number.

DARKENING THE BUBBLES ON THE ORS :

10. Use a **BLACK BALL POINT PEN** to darken the bubbles on the ORS.
11. Darken the bubble  **COMPLETELY.**
12. The correct way of darkening a bubble is as : 
13. The ORS is machine-gradable. Ensure that the bubbles are darkened in the correct way.
14. Darken the bubbles **ONLY IF** you are sure of the answer. There is **NO WAY** to erase or "un-darken" a darkened bubble.
15. Take **$g = 10 \text{ m/s}^2$** unless otherwise stated.

QUESTION PAPER FORMAT :

16. The question paper has three parts : Physics, Chemistry and Mathematics.

DO NOT BREAK THE SEALS WITHOUT BEING INSTRUCTED TO DO SO BY THE INVIGILATOR

Please see the last page of this booklet for rest of the instructions

SOME USEFUL CONSTANTS

Atomic No. : H = 1, B = 5, C = 6, N = 7, O = 8, F = 9, Al = 13, P = 15, S = 16, Cl = 17, Br = 35, Xe = 54, Ce = 58

Atomic masses : H = 1, Li = 7, B = 11, C = 12, N = 14, O = 16, F = 19, Na = 23, Mg = 24, Al = 27, P = 31, S = 32, Cl = 35.5, Ca = 40, Fe = 56, Br = 80, I = 127, Xe = 131, Ba = 137, Ce = 140

- | | |
|------------------------------------|--|
| • Boltzmann constant | $k = 1.38 \times 10^{-23} \text{ JK}^{-1}$ |
| • Coulomb's law constant | $\frac{1}{4\pi\epsilon_0} = 9 \times 10^9$ |
| • Universal gravitational constant | $G = 6.67259 \times 10^{-11} \text{ N-m}^2 \text{ kg}^{-2}$ |
| • Speed of light in vacuum | $c = 3 \times 10^8 \text{ ms}^{-1}$ |
| • Stefan-Boltzmann constant | $\sigma = 5.67 \times 10^{-8} \text{ Wm}^{-2}\text{-K}^{-4}$ |
| • Wien's displacement law constant | $b = 2.89 \times 10^{-3} \text{ m-K}$ |
| • Permeability of vacuum | $\mu_0 = 4\pi \times 10^{-7} \text{ NA}^{-2}$ |
| • Permittivity of vacuum | $\epsilon_0 = \frac{1}{\mu_0 c^2}$ |
| • Planck constant | $h = 6.63 \times 10^{-34} \text{ J-s}$ |

Space for Rough Work

HAVE CONTROL → HAVE PATIENCE → HAVE CONFIDENCE ⇒ 100% SUCCESS

BEWARE OF NEGATIVE MARKING

PART-1 : PHYSICS

SECTION-I : (Maximum Marks: 24)

- This section contains **SIX (06)** questions.
- Each question has **FOUR** options. **ONE OR MORE THAN ONE** of these four option(s) is (are) correct answer(s).
- For each question, choose the option(s) corresponding to (all) the correct answer(s)
- Answer to each question will be evaluated according to the following marking scheme:

Full Marks : +4 If only (all) the correct option(s) is (are) chosen.

Partial Marks : +3 If all the four options are correct but **ONLY** three options are chosen.

Partial Marks : +2 If three or more options are correct but **ONLY** two options are chosen and both of which are correct.

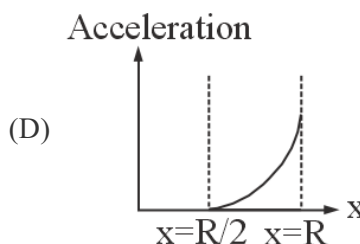
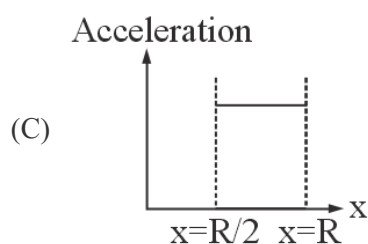
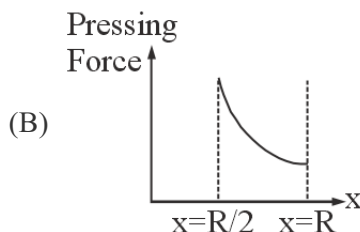
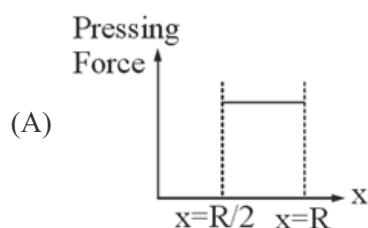
Partial Marks : +1 If two or more options are correct but **ONLY** one option is chosen and it is a correct option.

Zero Marks : 0 If none of the options is chosen (i.e. the question is unanswered).

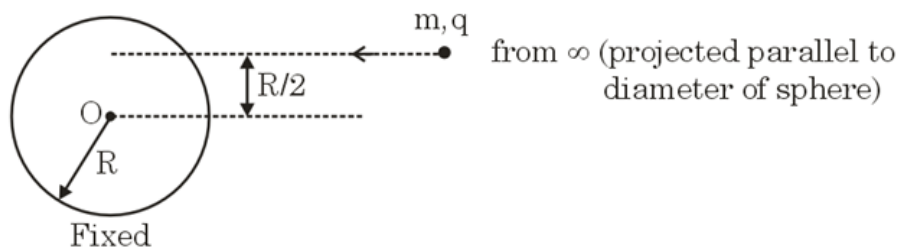
Negative Marks : -2 In all other cases.

- **For Example :** If first, third and fourth are the **ONLY** three correct options for a question with second option being an incorrect option; selecting only all the three correct options will result in +4 marks. Selecting only two of the three correct options (e.g. the first and fourth options), without selecting any incorrect option (second option in this case), will result in +2 marks. Selecting only one of the three correct options (either first or third or fourth option), without selecting any incorrect option (second option in this case), will result in +1 marks. Selecting any incorrect option(s) (second option in this case), with or without selection of any correct option(s) will result in -2 marks.

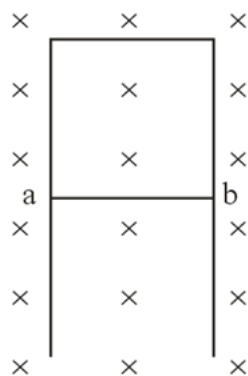
1. A tunnel is dug along a chord of the earth at a perpendicular distance $R/2$ from the earth's centre. The wall of the tunnel may be assumed to be frictionless. A particle is released from one end of the tunnel. The pressing force by the particle on the wall and the acceleration of the particle varies with x (distance of the particle from the centre) according to:



2. A particle of mass m kg and charge q coulomb is projected (from infinity) towards a non-conducting fixed sphere having the same charge uniformly distributed throughout volume of sphere. (see diagram). Find correct alternative(s).



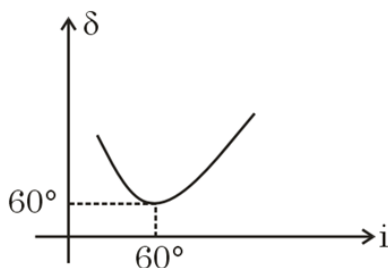
- (A) If charged particle is given velocity $\sqrt{\frac{4kq^2}{3mR}}$, it will just graze the solid sphere.
- (B) If charged particle is given velocity $\frac{8}{3} \sqrt{\frac{kq^2}{mR}}$, it will collide with the solid sphere, at angle 30° with normal to sphere surface.
- (C) Angular momentum of charged particle, about 'O', will remain conserved during motion.
- (D) Total energy of system remains conserved.
3. A copper wire ab of length ℓ , resistance r and mass m starts sliding under gravity at $t = 0$ down a smooth, vertical, thick pair of connected conducting rails as shown in figure. A uniform magnetic field B exists in the space in a direction perpendicular to the plane of the rails. Which option(s) is/are correct?



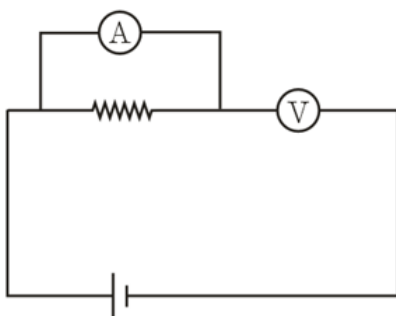
- (A) The displacement of the wire as a function of time is $v_m t - \frac{v_m^2}{g} (1 - e^{-gt/v_m})$. (where $v_m = \frac{mgr}{B^2 \ell^2}$)
- (B) The velocity of the wire as a function of time is $v_m (1 - e^{-gt/v_m})$. (where $v_m = \frac{mgr}{2B^2 \ell^2}$)
- (C) The downward acceleration of the wire at this instant is $g - \frac{B^2 \ell^2}{mr} v$.
- (D) The magnitude and direction of the induced current in the wire when speed of the wire is v is $\frac{vB\ell}{r}$, b to a

Space for Rough Work

4. For an equilateral glass prism, graph of deviation δ versus angle of incidence i is shown in figure. Choose incorrect option(s) :-



- (A) Refraction index of prism is $\sqrt{\frac{3}{2}}$.
- (B) For normal incidence on prism, total deviation suffered by ray is 60° .
- (C) Maximum possible deviation produced by prism is $30^\circ + \sin^{-1} \left(\sqrt{\frac{3}{2}} - \frac{1}{2} \right)$.
- (D) Maximum possible deviation produced by prism is $60^\circ + \sin^{-1} \left(\sqrt{\frac{3}{2}} - \frac{1}{2} \right)$.
5. By mistake, a voltmeter is placed in series and an ammeter in parallel with resistance in an electric circuit, with a cell in series.

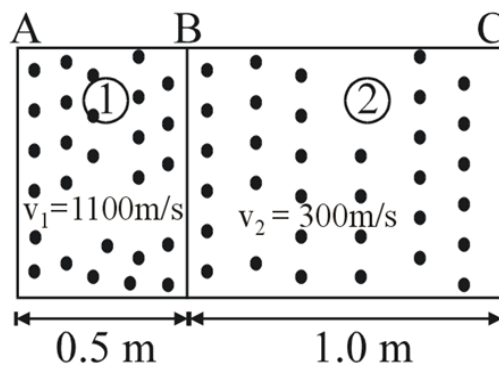


- (A) The devices may get damaged if emf of the cell is very high and the meters are non ideal.
- (B) If the devices (including battery) are ideal, ammeter will read zero current and voltmeter will read the emf of cell
- (C) If the devices are ideal, a large current will flow through the ammeter and it will be damaged
- (D) The main current in the circuit will be very low and almost all current will flow through the ammeter, if resistance of ammeter is much smaller than the resistance in parallel.

Space for Rough Work

6. A cylinder ABC consists of two chambers 1 and 2 which contains two different gases. The wall C is rigid but the walls A and B are thin diaphragms. A vibrating tuning fork approaches the wall A with velocity $u = 30 \text{ m/s}$ and air columns in chamber 1 and 2 vibrates with minimum frequency such that there is node (displacement) at B and antinode (displacement) at A.

(Assume velocity of sound in the first and second chamber be 1100 m/s and 300 m/s respectively. Velocity of sound in air 330 m/s .)



- (A) The fundamental frequency of air column is 1400 Hz .
 (B) The fundamental frequency of air column is 1650 Hz .
 (C) The frequency of tuning fork is 1500 Hz .
 (D) The frequency of tuning fork is 1200 Hz .

Space for Rough Work

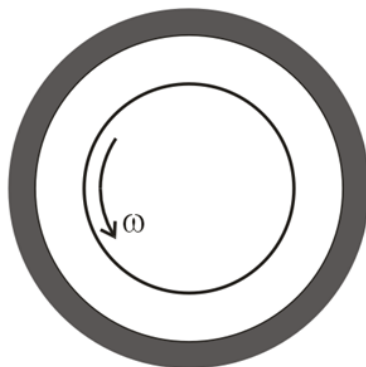
SECTION-II : (Maximum Marks: 24)

- This section contains **SIX (06)** questions. The answer to each question is a **NUMERICAL VALUE**.
- For each question, enter the correct numerical value of the answer in the place designated to enter the answer. If the numerical value has more than two decimal places, **truncate/round-off** the value to **Two** decimal places; e.g. 6.25, 7.00, -0.33, -.30, 30.27, -127.30, if answer is 11.36777..... then both 11.36 and 11.37 will be correct)
- Answer to each question will be evaluated according to the following marking scheme:

Full Marks : +4 If ONLY the correct numerical value is entered.

Zero Marks : 0 In all other cases.

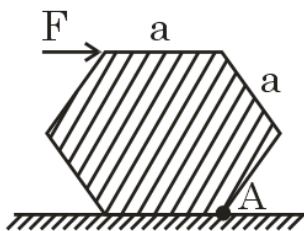
1. When 0.98 m long metallic wire is stretched, an extension of 0.02m is produced. An organ pipe 0.5m long & open at both ends, when sounded with this stressed metallic wire, produces 8 beats in its fundamental mode of both the instruments. By decreasing the strain in the wire, the number of beats are found to decrease. If Young's modulus of the wire is $z \times 10^{11} \text{ N/m}^2$, then find the value of z. (The density of metallic wire is 10^4 kgm^{-3} & sound velocity in air is 292 ms^{-1})
2. A cylinder of 150 mm radius rotates concentrically inside a fixed cylinder of 155 mm radius. Both cylinders are 300 mm long. Determine the viscosity of the liquid (in N-sec/m^2) which fills the space between the cylinders if a torque of 0.98 N-m is required to maintain an angular velocity of 60 r.p.m. (Given $\ln\left(\frac{0.155}{0.15}\right) = 0.328$).



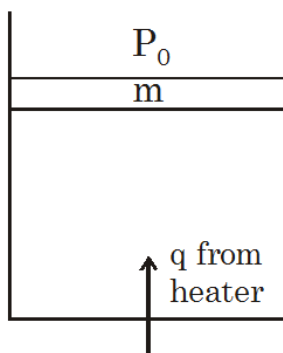
3. Consider a uniformly charged very long solid non-conducting cylinder of radius R and having volume charge density $\rho = \epsilon_0$. Find electric flux (in SI unit) through a gaussian spherical surface of radius 2R and having one of its diameter along the axis of cylinder and centre on axis of cylinder. Take : $R = \left(\frac{1}{\pi}\right)^{1/3}$

Space for Rough Work

4. A regular hexagonal uniform block of mass $m = 4\sqrt{3}$ kg rests on a rough horizontal surface with coefficient of friction μ as shown in figure. A constant horizontal force is applied on the block as shown. If the coefficient of friction is sufficient to prevent slipping before toppling, then the minimum force (in N) required to topple the block about its corner A is: ($g = 10 \text{ m/s}^2$)



5. An electron is shot into one end of solenoid. As it enters the uniform magnetic field within the solenoid, its speed is 800 m/s and its velocity vector makes an angle of 30° with the central axis of the solenoid. The solenoid carries 4.0 A current and has 8000 turn along its length. The number of revolutions made by the electron within the solenoid by the time it emerges from the solenoid's opposite end is $y \times 10^6$. Find value of y . (Use charge to mass ratio e/m for electron $= \sqrt{3} \times 10^{11} \text{ C/kg}$).
6. Two moles of an ideal monoatomic gas are contained in a vertical cylinder of cross sectional area A as shown in the figure. The piston is frictionless and has a mass m . At a certain instant, a heater starts supplying heat to the gas at a constant rate $q \text{ J/s}$. If the steady velocity of the piston under isobaric condition is given by $\frac{xq}{y(mg + P_0A)}$, then value of $\frac{x}{y}$ is : (All the boundaries are thermally insulated).



Space for Rough Work

SECTION-III : (Maximum Marks: 18)

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Zero Marks : 0 If no answer is given.

Negative Marks : -1 In all other cases

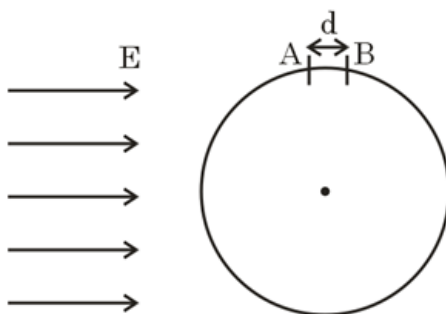
1. Figure shows two coherent sources S_1, S_2 vibrating in same phase. AB is irregular wire lying at a far distance from the sources S_1 and S_2 . Let $\frac{\lambda}{d} = 10^{-3}$, $\angle BOA = 0.12^\circ$. How many bright spots will be seen on the wire, including points A and B.



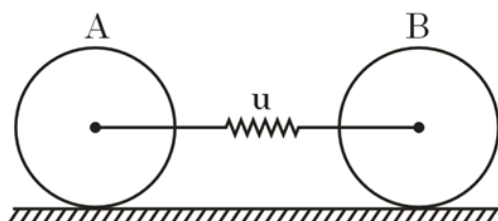
2. A radionuclide with half life $T = 693.1$ days emits β -particles of average kinetic energy $E = 8.4 \times 10^{-14}$ joule. This radionuclide is used as source in a machine which generates electrical energy with efficiency $\eta = 12.6\%$. Number of moles of the nuclide required to generate electrical energy at an initial rate of 441 KW is $n \times 10^m$ [where $n, m \in (1, 9)$] then find out value of $\frac{n}{m}$. ($\log_e 2 = 0.6931$, $N_A = 6 \times 10^{23}$)
3. A boy runs on a circular path of radius $R = 28$ m with a constant speed $u = 4$ m/s. Another boy starts from the centre of the path to catch the first boy. The second boy always remains on the radius connecting the centre of the circle and the first boy and maintains magnitude of his velocity constant $V = 4$ m/s. If the time of chase is $(10 + x)$ sec then $x = \dots\dots\dots$; $\left(\pi = \frac{22}{7}\right)$.

Space for Rough Work

4. Consider a point source emitting α -particles and receptor of area 1 cm^2 placed 1 m away from source. Receptor records any α -particle falling on it. If the source contains $N_0 = 3.0 \times 10^{16}$ active nuclei and the receptor records a rate of $A = 50000$ counts/second. Assume that the source emits alpha particles uniformly in all directions and the alpha particles fall nearly normally on the window. If decay constant is nearly $3n \times 10^{-(n+1)}$, then find the value of n .
5. A uniform insulating ring of mass m and radius R has total charge q distributed uniformly along its circumference except on length d ($d \ll R$). It is placed on smooth surface in uniform electric field parallel to length d which is between A & B. Initially ring is released from rest. If Maximum angular speed of ring in subsequent motion is $\frac{x}{R} \sqrt{\frac{qdE}{y\pi m}}$ [where x & y are minimum possible integer], then value of $\frac{y}{x}$ is:



6. Two spheres A and B of the same mass m and the same radius are placed on a rough horizontal surface. A is a uniform hollow sphere and B is uniform solid sphere. A and B can roll without sliding on the floor. They are also tied centrally to a light spring of spring constant k as shown in figure. A and B are released when the extension in the spring is x_0 . The amplitude of SHM of the sphere B is $\frac{5yx_0}{46}$. Find then value of y is:



Space for Rough Work

PART-2 : CHEMISTRY
SECTION-I : (Maximum Marks: 24)

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Partial Marks : +3 If all the four options are correct but **ONLY** three options are chosen.

Partial Marks : +2 If three or more options are correct but **ONLY** two options are chosen and both of which are correct.

Partial Marks : +1 If two or more options are correct but **ONLY** one option is chosen and it is a correct option.

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- **For Example** : If first, third and fourth are the **ONLY** three correct options for a question with second option being an incorrect option; selecting only all the three correct options will result in +4 marks. Selecting only two of the three correct options (e.g. the first and fourth options), without selecting any incorrect option (second option in this case), will result in +2 marks. Selecting only one of the three correct options (either first or third or fourth option), without selecting any incorrect option (second option in this case), will result in +1 marks. Selecting any incorrect option(s) (second option in this case), with or without selection of any correct option(s) will result in -2 marks.

1. Which of the following is/are **NOT** correct for N_2O –
 - (A) It is called laughing gas and it is produced by thermal decomposition of ammonium nitrate.
 - (B) It is nitrous oxide and it is produced by thermal decomposition of ammonium nitrite.
 - (C) It is not a linear molecule and it is produced by thermal decomposition of ammonium nitrate.
 - (D) It is least reactive in all oxides of nitrogen and it is produced by thermal decomposition of ammonium nitrite.
2. When MnO_2 is fused with KOH , a green coloured compound is formed. Choose correct statements about green coloured compound

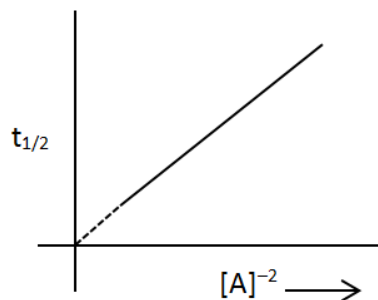
(A) It disproportionates in acidic medium	(B) It is paramagnetic in nature
(C) Geometry is tetrahedral	(D) It uses non axial d-orbitals in hybridisation

Space for Rough Work

3. For a reaction $2A + B \rightarrow C$, following information is known. Identify the options which is/are correct ?

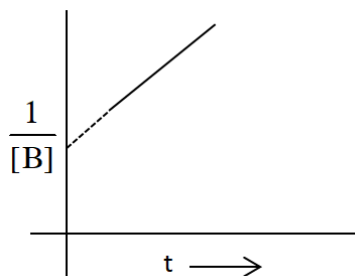
Information-1 :

When B is taken in very large amount, following graph was obtained :



Information-2 :

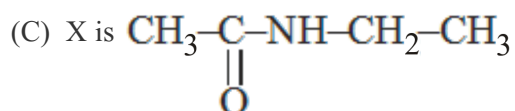
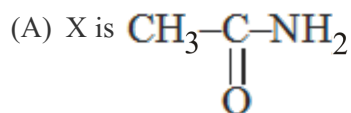
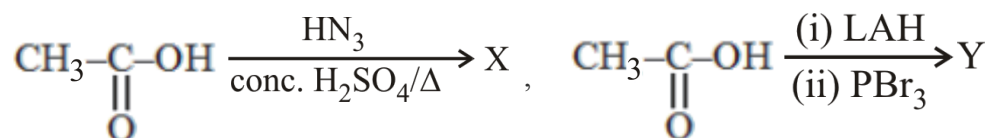
When $B = 1\text{ M}$ and $A = 2\text{ M}$ are mixed and graph of $\frac{1}{[B]}$ vs time is plotted, following graph is obtained :



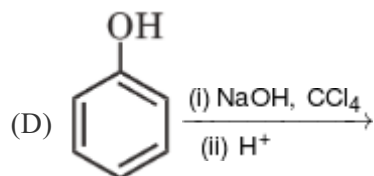
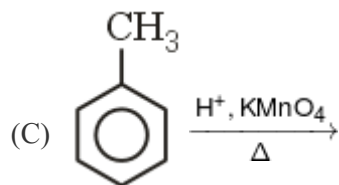
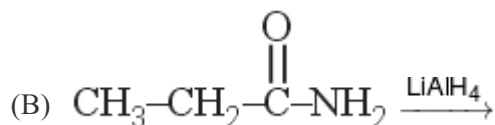
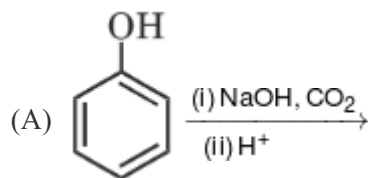
- (A) Order with respect to A is 2
 (B) Order with respect to B is 2
 (C) Overall order is 2
 (D) Order with respect to one of the reactant is negative.
4. Which of the following statement(s) is/are correct ?
- (A) Adsorption of gas on a solid surface is an entropy driven process.
 (B) A promoter when taken without catalyst can not change rate of reaction.
 (C) Muddy water can be coagulated more effectively by alum as compared to NaCl.
 (D) When excess AgNO_3 solution is added to KI solution and then resultant colloidal solution is subjected to electrophoresis, coagulation will be observed near anode.

Space for Rough Work

5. Choose the correct option(s) :



6. Which of the following has $-\text{COOH}$ group in major product ?



Space for Rough Work

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1. Assuming ideal geometries without any distortions, find value of $\left(\frac{P}{Q}\right)$ on basis of following information.

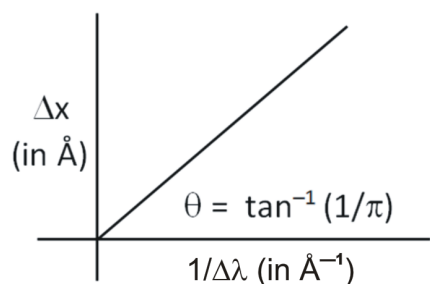
Properties & Features of Osmium complexes	$\text{OsCl}_x(\text{CO})_{2x}$	$\text{Os}(\text{CO})_{x+3}$
Net dipole moment	Non-zero	Zero
C–Os–C bond angles 90° & 180°	Yes	Yes
Geometrical isomerism possible	Yes	No
Maximum number of atoms in single plane	"P"	"Q"

2. Halide of Phosphorus PX_a is found to exist in ionic form in solid state, undergoes complete hydrolysis to give two acids H_bPO_c and HX . Salt NaX on reaction with conc H_2SO_4 liberates gas X_2 . Elemental form of X_2 at room temperature is liquid. Value of $\left(\frac{\text{at. wt. of X (in amu)}}{a + 2b + 3c}\right)$
3. Some amount of diatomic vander waal's gas is kept in a rigid container of volume 20 litre and subjected to change in temperature from 300K to 400 K. If 30 kcal heat is required and enthalpy change of process is observed to be $\frac{310}{9}$ kcal, then estimate the number of moles of the gas.

[Given : $b = 0.1 \text{ L mol}^{-1}$, $a = 1.25 \frac{\text{atm} - \text{L}^2}{\text{mol}^2}$]

Space for Rough Work

4. Calculate voltage (in volts) through which electron is accelerated if following plot is given for the electron?

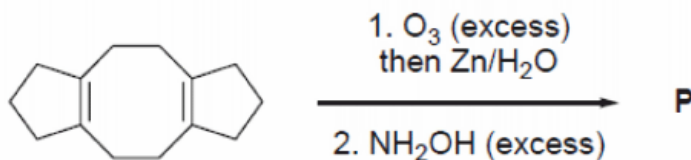


where Δx = uncertainty in position of electron

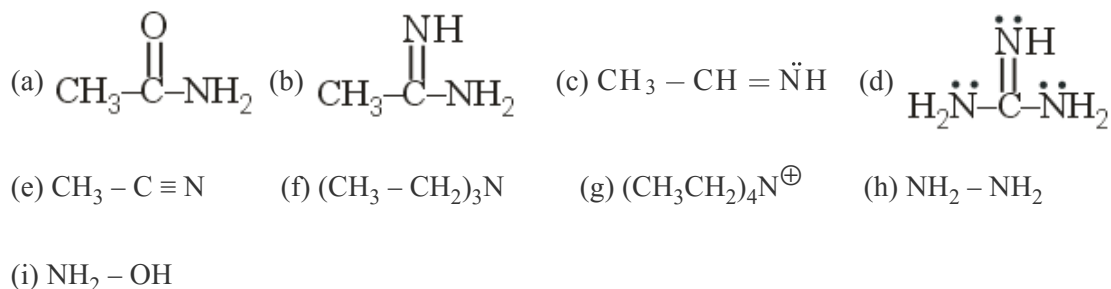
$\Delta \lambda$ = uncertainty in wavelength of electron

[Given : de-Broglie wavelength of electron, $\lambda = \sqrt{\frac{150}{V}} \text{ Å}$ where, V = accelerating potential]

5. In the reaction given below, the total number of atoms having sp^2 hybridization in the major product P is ____.



6. How many molecules are more basic than ethylamine (in gaseous state) ?



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- For each question, marks will be awarded in one of the following categories :

Full Marks : +3 If only the correct answer is given.

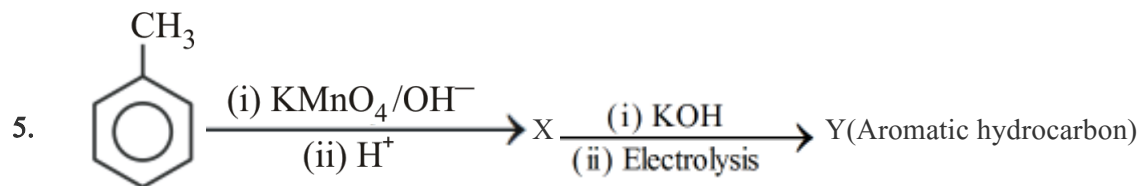
Zero Marks : 0 If no answer is given.

Negative Marks : -1 In all other cases

1. How many black colour precipitate from given below :-
 Ag_2S , CdS , HgS , BaSO_4 , $\text{Cu}(\text{OH})_2$, AgBr , NiS
2. Among the following, total number of compound(s) has/have more boiling point than He.
 H_2O_2 , H_2O , $\text{CH}_3\text{CH}_2\text{OH}$, CH_3OCH_3 , T_2 , D_2 , H_2
3. 100 ml of 0.1 M H_3PO_4 is titrated with 0.05 M NaOH solution till 2nd equivalence point. Then, resultant solution was mixed with 10 mL of 0.5 M HCl solution. pH after addition of HCl will be-
 [For H_3PO_4 $K_{a_1} = 10^{-3}$, $K_{a_2} = 10^{-8}$, $K_{a_3} = 10^{-13}$]

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4. A 1.1 gm sample of copper ore is dissolved and Cu^{2+} (aq.) is titrated with excess KI. The liberated I_2 requires 12.12 mL of 0.1M $\text{Na}_2\text{S}_2\text{O}_3$ solution for titration. What is the % of copper by mass in the ore. (in nearest integer) (Atomic weight of Cu = 63.5)

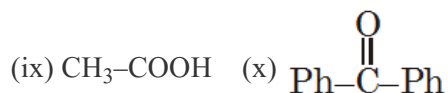
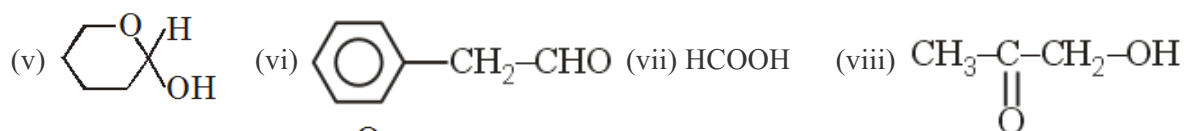
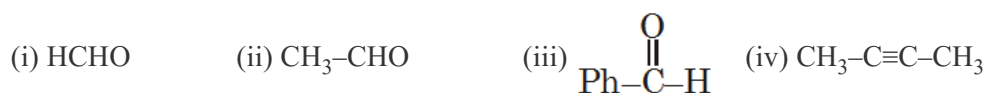


In 'Y' total number of sigma bond formed by overlapping of $(\text{sp}^2 - \text{sp}^2)$ orbital(s) = (P)

Total number of πe^- = (Q)

value of P – Q is

6. Identify total number of compounds which give both tollen's and fehling test ?



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PART-3 : MATHEMATICS
SECTION-I : (Maximum Marks: 24)

- This section contains **SIX (06)** questions.
- Each question has **FOUR** options. **ONE OR MORE THAN ONE** of these four option(s) is (are) correct answer(s).
- For each question, choose the option(s) corresponding to (all) the correct answer(s)
- Answer to each question will be evaluated according to the following marking scheme:

Full Marks : +4 If only (all) the correct option(s) is (are) chosen.

Partial Marks : +3 If all the four options are correct but **ONLY** three options are chosen.

Partial Marks : +2 If three or more options are correct but **ONLY** two options are chosen and both of which are correct.

Partial Marks : +1 If two or more options are correct but **ONLY** one option is chosen and it is a correct option.

Zero Marks : 0 If none of the options is chosen (i.e. the question is unanswered).

Negative Marks : -2 In all other cases.

- **For Example :** If first, third and fourth are the **ONLY** three correct options for a question with second option being an incorrect option; selecting only all the three correct options will result in +4 marks. Selecting only two of the three correct options (e.g. the first and fourth options), without selecting any incorrect option (second option in this case), will result in +2 marks. Selecting only one of the three correct options (either first or third or fourth option), without selecting any incorrect option (second option in this case), will result in +1 marks. Selecting any incorrect option(s) (second option in this case), with or without selection of any correct option(s) will result in -2 marks.

- Let $g(x) = ax^2 + bx + c$, $a, b, c \in \mathbb{N}$ and $\int_0^1 g(x) dx = \frac{11}{6}$. Let $f(x)$ be a continuous and derivable function in (x_1, x_2) . If $f(x) \cdot f'(x) \geq x\sqrt{1 - (f(x))^4}$ and $\lim_{x \rightarrow x_1^+} (f(x))^2 = 1$ and $\lim_{x \rightarrow x_2^-} (f(x))^2 = \frac{1}{2}$, then minimum value of $[x_1^2 - x_2^2]$ is equal to, where $[.]$ denotes greatest integer function.
(A) $2 - abc$ (B) $a + b - c$ (C) $b - c$ (D) $a - b$
- PQ and RS are normal chord to parabola $y^2 = 8x$ at P and R on the curve respectively and the points P, Q, R, S are concyclic. If A is vertex of parabola, then
(A) centroid of $\triangle APR$ lies on x-axis (B) centroid of $\triangle APR$ lies on y-axis
(C) PR is parallel to tangent at the vertex (D) PR is parallel to axis of parabola

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3. If a set 'A' is $\{1, 0, -1, 2\}$ then which of the following is(are) correct ?
- (A) A square matrix of order 3 is constructed whose elements are from set 'A' then the probability that matrix is skew symmetric matrix is $\frac{27}{4^9}$.
- (B) A square matrix of order 3 is constructed whose elements are from set 'A' then probability that matrix is symmetric is $\frac{1}{64}$.
- (C) A function $f : A \rightarrow A$ is defined then probability that function is onto is $\frac{3}{32}$.
- (D) A function $f : A \rightarrow A$ is defined and f is onto then the probability that $f(i) \neq i \forall i \in A$ is $\frac{3}{8}$.
4. Let $P_k (k = 1, 2, 3, \dots, n)$ be the n^{th} root of unity. Let $z = a + ib$ (where a & b are positive real number) and $A_k = \text{Re}(z) \text{Re}(P_k) + i \text{Im}(z) \text{Im}(P_k)$ then which of the following is/are true :
- (A) A_k lies on ellipse
- (B) A_k lies on hyperbola
- (C) If S be the focus of trajectory on which A_k lies then $\sum_{k=1}^n A_k S = na$
(where $A_k S$ denotes distance between A_k & S)
- (D) If S be the focus of locus of A_k the $\sum_{k=1}^n (A_k S)^2 = \frac{n}{2} (3a^2 - b^2)$
(where $A_k S$ denotes distance between A_k & S)

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5. Let $I = \int_1^2 \frac{\sin x}{x^2} dx$, then

(A) $I < \ln 2$

(B) $I > \ln 2 - \frac{1}{4}$

(C) $I > \frac{1}{2\sqrt{2}}$

(D) $I < \frac{2}{\pi} \left[\ln \left(\frac{\pi^2}{2} \right) + \left(\frac{2-\pi}{2} \right) \right]$

6. The position vector of the vertices A, B & C of a tetrahedron ABCD are (1, 1, 1), (1, 0, 0) & (3, 0, 0) respectively. The altitude from the vertex D to the opposite face ABC meets the median through A of $\triangle ABC$ at point E. If AD = 4 units and volume of tetrahedron = $\frac{2\sqrt{2}}{3}$, then the correct statement(s) among the following is/are :

(A) The altitude from vertex D = 2 units

(B) There is only one possible position for point E

(C) There are two possible positions for point E

(D) vector $\hat{j} - \hat{k}$ is normal to the plane ABC

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SECTION-II : (Maximum Marks: 24)

- This section contains **SIX (06)** questions. The answer to each question is a **NUMERICAL VALUE**.
- For each question, enter the correct numerical value of the answer in the place designated to enter the answer. If the numerical value has more than two decimal places, **truncate/round-off** the value to **Two** decimal places; e.g. 6.25, 7.00, -0.33, -0.30, 30.27, -127.30, if answer is 11.36777..... then both 11.36 and 11.37 will be correct)
- Answer to each question will be evaluated according to the following marking scheme:

Full Marks : +4 If ONLY the correct numerical value is entered.

Zero Marks : 0 In all other cases.

- Let $f: \mathbb{R}^+ \rightarrow \mathbb{R}$ be a differentiable function satisfying $f(x) = e + (1-x) \ln\left(\frac{x}{e}\right) + \int_1^x f(t) dt \forall x \in \mathbb{R}^+$. If the area enclosed by the curve $g(x) = x[f(x) - e^x]$ lying in the fourth quadrant is A, then the value of A^{-2} is
- $\vec{p} = 3^{(\sin x + \tan x)} \hat{i} + 3^{(\tan x - 1)} \hat{j} - 3^{\sin x} \hat{k}$ and $\vec{q} = 3^{(\sin x - 1)} \hat{i} + 3^{(\sin x + \tan x)} \hat{j} + (3^{\tan x}) \hat{k}$. If sum of all values of x in $[0, 50\pi]$ for which $\vec{p} \cdot \vec{q}$ is minimum is S, then find the value of $\left(\frac{S}{255\pi}\right)$
- Number of points where $f(x) = [[x] + |x + \sin x|] \operatorname{sgn}(x^2 - 1)$, $x \in (-6, 6)$ is discontinuous is _____ (where $[.]$ denotes greatest integer function and $\operatorname{sgn}(x)$ is signum function)
- Let $A = \begin{bmatrix} 2 & 1 \\ 1 & 0 \end{bmatrix}^n = (a_{ij}(n))$, (where $1 \leq i \leq 2$ & $1 \leq j \leq 2$), then the value of $\lim_{n \rightarrow \infty} \frac{a_{12}(n)}{a_{22}(n)}$ is equal to _____
- PQ and PR are two tangents to a parabola $y^2 = 4ax$ and tangent to parabola at a third point S cuts these tangents at A and B respectively then value of $\frac{PA}{PQ} + \frac{PB}{PR}$ is
- Let $f(x)$ be a differentiable function defined for all $x > 0$ and $\lim_{x \rightarrow 0} f(x) = 0$. Also $f(x)$ satisfies
 (i) $f(xy) = xf(y) + yf(x) \forall x, y \in \mathbb{R}^+$ (ii) $\int_0^x f(x) dx = \frac{f(x^2) - x^2}{4}$
 If the area enclosed by $y = f(x)$ lying in the fourth quadrant is S, then $\log_{\sqrt{6}} \left(\frac{9}{S}\right)$ is equal to

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SECTION-III : (Maximum Marks: 18)

- This section contains **SIX (06)** questions.
- The answer to each question is a **SINGLE DIGIT INTEGER** ranging from 0 to 9, both inclusive.
- For each question, enter the correct integer value of the answer in the place designated to enter the answer.
- For each question, marks will be awarded in one of the following categories :

Full Marks : +3 If only the correct answer is given.

Zero Marks : 0 If no answer is given.

Negative Marks : -1 In all other cases

1. The number of distinct real solutions of equation $4\sin^4 x + 4\sin^3 x + 3\sin^2 x = 1 + \sin x$, in the interval $[0, 2\pi]$, is
2. A closed rectangle box with square base has been made from certain solid material and is painted red from outside and blue from inside. Its inner volume is 512 cubic unit and all six walls is 2 unit thick. Now, some rectangular openings through the wall is carved out such that removed red painted area is half of the initial inner blue surface area of box. Then, the material volume of this partially closed box is minimum when height of the box is
3. If $T_n = \frac{(2n+1)(2n^2+2n-1)}{(n-1)^2 n^2 (n+1)^2 (n+2)^2}$, $n \in \mathbb{N}$, $n \geq 2$. If $K = \lim_{n \rightarrow \infty} \sum_{r=2}^n T_r$ then $27K$ is equal to
4. Let $z = \cos\left(\frac{2\pi}{2n+1}\right) + i \sin\left(\frac{2\pi}{2n+1}\right)$, where $n \in \mathbb{I}^+$ & $A = \sum_{r=1}^n z^{2r-1}$ & $B = \sum_{r=1}^n z^{2r}$.
If $ax^2 + bx + \sec^2\left(\frac{\pi}{2n+1}\right) = 0$ has roots A and B then find the value of $(a+b)$
5. Let $f(x)$ and $g(x)$ are two functions differentiable everywhere such that $g(x) = \frac{1}{f'(x) - 2g(x)(f(x))^2}$ and $f(x) = \frac{3}{g'(x) - 2f(x)(g(x))^2}$ where $f(0) = 1$, $g(0) = 1$. If $A = \int_{-\frac{\pi}{48}}^{\frac{\pi}{48}} f(x)g(x)dx$ then find the value of e^{8A}
6. In a ΔABC , AB is parallel to y-axis, BC is parallel to x-axis and centroid is at (2, 1). If median through C is $x - y = 1$ and distance of median through A from origin is ℓ , then the value of $\sqrt{17}\ell$ is

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NAME OF THE CANDIDATE

FORM NO.

I have read all the instructions
and shall abide by them.

Signature of the Candidate

I have verified the identity, name and Form
number of the candidate, and that question
paper and ORS codes are the same.

Signature of the Invigilator

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